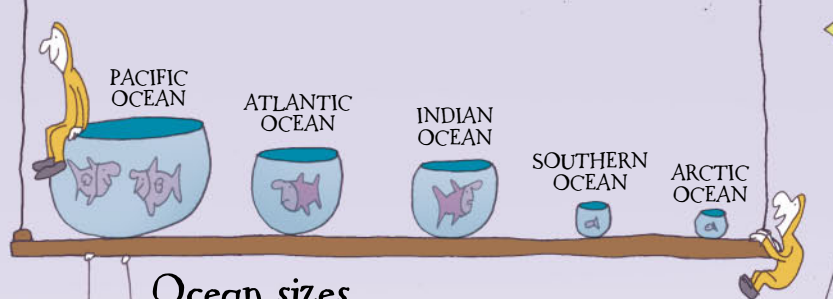


OCEANS

More than two-thirds of Earth is covered in saltwater, most of which is contained within five great oceans that are home to millions of different fish and marine mammals. Until recently humans knew little about this watery world, but new technology has opened it up to human exploration and discovery.



Ocean sizes

The world's five oceans and many seas occupy 139 million sq miles (361 million sq km) of Earth's surface. The Pacific Ocean is biggest in size, occupying a colossal 46.4 percent of the total watery surface of Earth.

This place is a dive.

Good reef!

Don't leave me on the shelf!

What a shelf life!

Continental shelf

The edge of each continent slopes down into the sea, forming a shelf of shallow water.

Tides

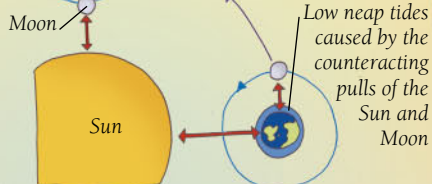
The gravitational pull between the spinning Earth, Moon, and Sun causes the oceans around the globe to rise and flood the shore, and then fall back, or ebb, again every 12 hours.

Path of Moon's rotation around Earth

High spring tides caused by the combined pull of the Sun and Moon

Path of Earth's rotation

Low neap tides caused by the counteracting pulls of the Sun and Moon



Ocean-basin floor

Past the continental shelf, the ocean-basin floor plunges to more than 6,560ft (2,000m) deep. Here, midocean ridges and trenches snake along the ocean floor. Descend even further to 16,400ft (5,000m) and discover the abyssal plain—the deepest, darkest part of the ocean floor, covered in a thick ooze.

Continental slope

Beyond the continental shelf the ocean floor plunges steeply down the continental slope toward the deepest part of the ocean.

It's a slippery slope.

Watch my nose dive.

Hello, sushi!

Midocean ridge

As the Earth's tectonic plates pull apart on the ocean floor, magma rises up from the Earth's interior to form a midocean ridge.

Pillow lava

When bubbles of lava erupt from a midocean ridge, they cool and solidify upon contact with the cold seawater, creating pillow-shaped lumps of rock that litter the ocean floor.



Coral reefs

These underwater tropical forests are formed from the hard outer skeletons of tiny sea creatures called polyps.

It's a current affair.

Red arrows: warm currents
Blue arrows: cold currents

South Pacific gyre Gulf Stream South Indian gyre

Surface currents

The world's winds disturb the surface of the oceans, causing currents that flow in curves, known as gyres, due to Earth's rotation. Warm currents flow away from the Equator; cold currents flow toward it.

New sea floor forms a midocean ridge.

Moving tectonic plate

Rising magma

Tectonic plates continue to pull apart.

Seafloor spreading

As two plates pull apart, magma rises up between them to form a ridge. This new seafloor is in turn pushed apart by more magma rising beneath it in a process called seafloor spreading, which is gradually widening the ocean.